Chapter 24 worked solutions

Speech and Audio

Exercises 1–4 work through audio frames, spectrogram axes, decoding tasks, and transcript errors by hand. Exercises 9–11 design evaluations for noisy recordings, clips without speech, and live captions.

1. Count complete frames

Number the recording's samples from 0 through 23. A frame takes eight samples, including its starting sample, so a frame starting at s ends at $s + 7$. The last possible start is 16 because that frame ends at 23. Begin at 0 and move forward by the hop each time, stopping once the next start would exceed 16.

Hop (samples)	Frame start indices	Sample ranges, inclusive
4	0, 4, 8, 12, 16	0–7, 4–11, 8–15, 12–19, 16–23
8	0, 8, 16	0–7, 8–15, 16–23

With hop four, the first pair shares samples 4, 5, 6, and 7. Every consecutive pair shares **four samples**, since each move advances by four of the eight samples. With hop eight, the next frame starts immediately after the previous frame ends, so consecutive frames share **zero samples**.

2. Read the axes

Look at a single 16-sample frame first. A frequency bin is one of the frequencies the Fourier transform checks. Bin 0 measures the constant component, and bin 1 checks for one cycle per frame. A frame spans $16/8,000 = 0.002$ seconds, so an oscillation that completes one cycle in that time completes 500 cycles in a second. Bin k checks for k cycles in the same frame. The frequency spacing is therefore $\Delta f = f_s/N = 500\,Hz$.

Bin k	Frequency $k\Delta f$ (Hz)
0	0
1	500
8	4,000

Now look across frames. The hop advances the next start by eight sample intervals. At $8,000$ samples per second, the time between starts is

$$\Delta t = \frac{H}{f_s} = \frac{8}{8,000}\,s = 0.001\,s = 1\,ms.$$

At a fixed sample rate, N sets the **frequency spacing** within each frame. A longer frame brings the bins closer together. H sets the **time spacing** between frame starts. A larger hop puts those starts farther apart.

3. Choose a decoding task

The speaker says *buenos días* in both cases. In Whisper's decoder settings, the language identifies the language spoken in the recording. Choose Spanish for both requests, then use the task setting to select the output.

Requested result	Spoken language setting	Task setting	Example text
Spanish transcript	Spanish (`es`)	`transcribe`	*Buenos días*
English translation	Spanish (`es`)	`translate`	*Good morning*

Both requests supply the same clip and use the same audio preprocessing, so the encoder receives the same log-Mel features, the frequency summaries computed for each frame of the recording. The encoder can produce one acoustic representation for both requests. The decoder reads that representation and uses the task token to decide whether to write the words in Spanish or render their meaning in English.

4. Align the words

Start by keeping the words that agree in order. Replace *blue* with *red* and add *now* after *door*. Read down the alignment to follow both sentences in their original order. A gap in the reference column marks a hypothesis word inserted there.

Reference	Hypothesis	Operation
turn	turn	Match
at	at	Match
the	the	Match
blue	red	Substitute
door	door	Match
—	now	Insert

This uses two edits. The reference has five words and the hypothesis has six, so a one-edit solution would have to insert a word. An insertion leaves all five original words in place, including *blue*, which is absent from the hypothesis. That rules out a one-edit solution, so the displayed alignment is minimum.

Word error rate (WER) divides the number of edits by the number of reference words. The counts here are one substitution, zero deletions, and one insertion.

$$WER = \frac{1 + 0 + 1}{5} = \frac{2}{5} = 40\%.$$

If the sentences swap roles, *turn at the red door now* becomes the six-word reference. Reverse the edits by replacing *red* with *blue* and deleting *now*. Each edit can be undone with one edit, so the minimum total is still two. The new denominator is **six reference words**, which gives $2/6 = 1/3 \approx 33.3\%$.

9. Check noise without leakage

Suppose a speaker records *please turn left*. The clean file and a copy mixed with street noise contain the same words, voice, and timing. Putting the clean version in training and the noisy one in test would let the system train on the very utterance it is supposed to recognize under noise. Holding out one recording is also insufficient if the goal is recognition of new speakers.

Assign complete speaker groups to the splits. For example, use 18 speakers for training, 6 for validation, and 6 for test. If one speaker made several recordings, keep all of them in that speaker's split. Attach the original recording ID and condition to each file and keep the clean and noisy versions together. For instance, both copies of every recording by a held-out speaker belong in test. Verify that no speaker ID or original recording ID appears in more than one split. If a recording contains multiple speakers, keep linked speakers and recordings together when choosing the groups.

Choose preprocessing, decoding options, and any noise training settings on validation. Keep the held-out test settings fixed. For artificially added noise, record the noise source, its level relative to the speech, and how the mix was made. A claim about transfer to new noise sources needs an additional split by noise source. For the new-speaker comparison here, score the two versions against the same transcript using the same word normalization.

Report pooled WER on clean test clips and noisy test clips separately. Sum the edits within each condition and divide by its number of reference words. For example, if both subsets contain 200 reference words, 12 clean edits give 6 percent and 28 noisy edits give 14 percent. The noisy condition is 8 percentage points higher on this illustrative set. Report the word, clip, and speaker counts with both scores. If noise levels vary, show results by level as well. For uncertainty on the clean-to-noisy difference, resample speakers and keep each recording's two versions together. If speakers are linked by shared recordings, resample the linked groups instead. Multiple clips from a group do not provide independent speaker measurements.

10. Score false words

A captioner that writes *thank you* during a recording of a fan has inserted words even though nobody spoke. Build a no-speech set with both digital silence and realistic non-speech sound. Examples of the second group are a fan, a passing car, and footsteps. Listen to the clips to verify they contain no spoken words. Use comparable clip lengths or record duration, because a longer clip provides more opportunities to generate text.

Give each clip an empty reference word list. Apply the same text normalization and word tokenization used for the speech test. Remove output metadata such as timestamps and explicit no-speech markers before counting transcript words. Count any remaining word as a false word on this set. A silent clip with only a no-speech marker is a successful empty transcript.

Use the **false-word clip rate**, the number of no-speech clips with at least one transcript word divided by the number of no-speech clips. Report it separately for silence and for non-speech sound. If one of two fan clips produces *thank you* and the other produces nothing, the false-word clip rate for those two clips is $1/2$, and the total false-word count is two. Counting words as well as affected clips distinguishes an isolated guess from a long invented transcript. If durations vary, false words per minute provides a separate rate with time in the denominator.

Clip-level WER is undefined here because the reference word count is zero. The generated words still contribute insertions to a pool that also contains speech clips. For example, suppose the speech clips have 10 reference words and two edits, and the no-speech clips produce three false words. The combined WER is $(2 + 3)/10 = 0.5$. The spoken subset alone would give $2/10 = 0.2$. Each false word adds an edit to the numerator, and an empty reference adds no words to the denominator. Report which no-speech clips entered the pool, their insertions, and the speech-only score alongside the combined score.

11. Measure live captions

Suppose a speaker finishes saying *turn* at 1.20 seconds. One system first displays *turn* at 1.35, replaces it with *term* at 1.48, and shows *turn* again at 1.62. The second display of *turn* survives to the final caption. The word first appeared 150 ms after it was spoken and stabilized 420 ms after it was spoken. A single latency number would miss the revision in between.

Stream the same held-out recordings to both systems in real time. Use an annotated end time for each spoken word, obtained from a checked alignment of the reference words with the audio. Use a common monotonic clock to map each word's audio end time to elapsed wall time. Log the time each audio chunk is captured or released, the time it is delivered to the system, and the time of every displayed caption revision and finalization. Save the text and time of each revision. For a model-output measurement, log emission time and identify that boundary separately. Flush the stream and wait for the configured final output so late revisions are observed.

Match the displayed words to reference positions with a fixed alignment policy. Use reference timestamps to disambiguate repeated words when the system supplies word times. For untimed ties, choose the earliest plausible reference occurrence while preserving the order and identities of words already matched in the caption history. Review uncertain alignments consistently. For a reference position, first-appearance latency is the first time the correct word is displayed there minus its spoken end time. Stabilization latency is the earliest time that the correct word appears there and then remains correct through the final caption, minus the same end time. Track the logical caption history so normal scrolling does not erase a committed word. Count replacements and removals as revisions and track ordinary suffix appends separately. A system can begin displaying a word before it has finished in the audio, so retain signed latencies under this end-time convention.

A word that never appears correctly has no first-appearance time, and a word that fails to stay correct has no correct stabilization time. Report both coverage counts against all reference words. Give median and 95th-percentile latencies over the applicable words, revision counts, and final WER. Report the latency difference on reference positions with defined values for both systems alongside those coverage numbers. A fast system can otherwise look better by omitting the slowest words.

Record hardware, runtime, numerical precision, and how many streams ran concurrently. Also record sample rate, chunk size and cadence, queue and lookahead buffering, any voice-activity rule, and the update and commit policy. Keep the measurement setup comparable and document any buffering that differs between systems. The same offline WER can coexist with different delay, different revision frequency, or both.